

Blood Sampling for Nurses: Comprehensive MCQ and Practice Questions on Tubes, Plasma vs Serum, and Pre-analytical Errors

1. Choose the correct answers:

1. The main message of “Know Your Tubes and Colors” is that tube color indicates the:
 - a) Needle size
 - b) Patient diagnosis
 - c) Additive, sample type, and test type
 - d) Time of collection

2. The red-top tube (no additive or clot activator only) is mainly used for:
 - a) CBC and blood film
 - b) Glucose only
 - c) Coagulation profile
 - d) Many biochemistry, serology, immunology tests

3. The gold/yellow SST tube contains gel plus clot activator and is typically used for:
 - a) Routine chemistry and hormones
 - b) ESR only
 - c) Cross-matching
 - d) Arterial blood gases

4. The light-blue tube contains sodium citrate and provides:
 - a) Fluoride plasma
 - b) Heparinized plasma
 - c) Citrated plasma
 - d) Whole blood

5. Coagulation tests such as PT, PTT, and INR are mainly performed on samples from which tube?
 - a) Green
 - b) Light blue

- c) Lavender
 - d) Gray
6. The lavender/purple tube contains EDTA and is typically used for:
- a) Liver enzymes
 - b) CBC, blood film, some HbA1c
 - c) Lipid profile
 - d) Thyroid hormones only
7. The gray tube containing fluoride and oxalate is mainly used to measure:
- a) Coagulation profile
 - b) Thyroid hormones
 - c) Glucose (inhibits glycolysis)
 - d) ESR
8. Plasma can be defined as the liquid part of blood:
- a) After clotting, without clotting factors
 - b) With anticoagulant present, containing fibrinogen
 - c) After removal of buffy coat
 - d) Used only for transfusion
9. Serum differs from plasma because serum:
- a) Contains more fibrinogen
 - b) Requires an anticoagulant
 - c) Has clotting factors removed with the clot
 - d) Has higher cell content
10. Compared with serum, plasma usually has:
- a) Slower separation time
 - b) Faster separation time
 - c) No use in laboratories
 - d) No need for centrifugation

11. Whole blood is best described as:
- a) Serum without cells
 - b) Plasma plus cells (RBCs, WBCs, platelets)**
 - c) Buffy coat only
 - d) Red cells only
12. A major visible sign of hemolysis in a sample tube is:
- a) Clear, colorless supernatant
 - b) Red-colored serum or plasma**
 - c) Greenish plasma
 - d) Thick buffy coat
13. Delaying separation of whole blood for many hours mainly causes:
- a) Increase in glucose level
 - b) Decrease in glucose and changes in some analytes**
 - c) No change in any analyte
 - d) Instant clot formation in all tubes
14. Clotted anticoagulated samples are usually the result of:
- a) Using a large-bore needle
 - b) Inadequate mixing of EDTA, citrate, or heparin tubes**
 - c) Centrifuging too soon
 - d) Using serum tubes only
15. In the short scenario, drawing blood from an arm with IV dextrose into a gray tube mainly led to:
- a) Falsely low glucose and reliable potassium
 - b) Very high measured glucose and unreliable potassium because of hemolysis**
 - c) Normal glucose with normal potassium
 - d) Only a delay in transport time
16. Most laboratory errors occur in which phase of testing?
- a) Analytical
 - b) Post-analytical

c) Pre-analytical

d) Reporting

17. Which of the following is a common nursing error in blood sampling?

a) Correct patient identification

b) Using the correct tube color

c) Taking blood from the arm with running IV fluids

d) Gentle mixing of tubes

18. Underfilled light blue citrate tubes mainly affect:

a) Blood group tests

b) Coagulation tests

c) Urine analysis

d) Microbiology cultures

19. Which of the following sequences correctly summarizes the pre-analytical pathway “From vein to analyzer”?

a) Correct tube → patient ID → transport → venipuncture

b) Correct patient ID → correct tube → proper venipuncture → correct fill volume → gentle mixing → timely transport

c) Venipuncture → mixing → patient ID → transport → tube choice

d) Transport → analysis → reporting → patient ID

20. Hemolysis in serum/plasma is most commonly caused by:

a) Delayed centrifugation only

b) Rough handling, small needle, shaking tubes, long tourniquet time

c) Overfilling the tube

d) Low room temperature

2. Fill-in-the-blank

1. The red-top tube usually provides a **serum** sample used for many biochemistry, serology, and immunology tests.
2. The gold/yellow SST tube contains a **gel** plus clot activator to separate serum.
3. The additive in the light-blue tube is **sodium citrate**
4. Green tubes typically contain lithium or sodium **heparin** to produce heparinized plasma.
5. Lavender or purple tubes contain **EDTA** and are used mainly for CBC.
6. Underfilled or overfilled tubes lead to an incorrect blood to **additive** ratio, especially in light-blue tubes.
7. Plasma is the liquid part of blood with an **anticoagulant** present and still contains clotting factors including fibrinogen.
8. Serum is obtained after the blood **clots** and therefore lacks **clotting** factors.
9. The buffy coat in a centrifuged tube mainly contains white blood cells and **platelets**.
10. In hemolysis, red blood cells rupture and release their contents, giving the plasma or serum a **red** color.
11. Prolonged delay in separation allows cells to continue **glycolysis**, leading to a fall in glucose concentration.
12. Correct pre-analytical steps start with correct patient **ID** and choosing the correct **tube**.
13. In the clinical scenario, the nurse squeezed the patient's arm strongly, which contributed to **hemolysis** of the sample.
14. Because of the pre-analytical error in the scenario, the doctor delayed **insulin** adjustment for the diabetic patient.

15. The key message of the scenario is that “One small pre-analytical **mistake**; _ created a big **clinical** __ problem.”

Match table 1 – Tube color and additive

Column A	Column B
A. Red	1. Gel + clot activator
B. Gold / Yellow (SST)	2. Fluoride + oxalate
C. Light blue	3. Sodium citrate
D. Green	4. Lithium or sodium heparin
E. Gray	5. No additive or clot activator

Answers: A-5, B-1, C-3, D-4, E-2

Match table 2 – Concept and description

Column A	Column B
A. Plasma	1. Liquid after clotting, lacks fibrinogen.
B. Serum	2. Plasma plus cells (RBCs, WBCs, platelets).
C. Whole blood	3. Red-colored sample due to RBC destruction.
D. Hemolysis	4. Liquid part with anticoagulant, contains clotting factors.
E. Delay in separation	5. Leaving whole blood for hours causing glucose decrease.

Answers: A-4, B-1, C-2, D-3, E-5